Convergence and Taylor Series — Geometric Series, p-Series, and the nth-Term Test

Classifying geometric and p-series by their own parameters, and using the nth-term test to detect divergence

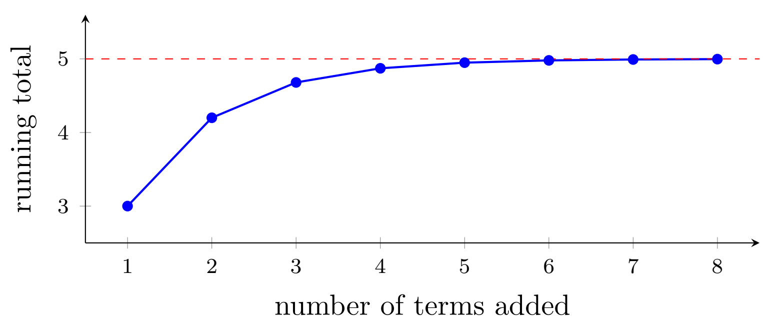
Directions:

- Name the *family* first (geometric? p-series? neither?), then name the parameter that decides it (r or p), then state the conclusion.
- A conclusion needs its reason attached. "Converges by the geometric series test, since $|r| = \frac{2}{5} < 1$" earns full credit; "converges" does not.
- Give exact values — fractions, not decimals — unless a problem says otherwise.
- Show the substitution into whichever formula you use.

Two pictures of "adding infinitely many terms"

Model A — partial sums of a geometric series. Each new term of $3 + \frac{6}{5} + \frac{12}{25} + \cdots$ is $\frac{2}{5}$ of the one before it. The running totals settle down:

k	1	2	3	4	5
S_k (running total)	3	4.2	4.68	4.872	4.9488



The running totals climb toward the dashed line and never pass it.

Model B — partial sums of the harmonic series $1 + \frac{1}{2} + \frac{1}{3} + \cdots$, whose terms also shrink to zero:

terms added	1	10	100	1000
running total	1	2.929	5.187	7.485

These running totals never settle. Shrinking terms are not enough — that is the whole point of problems 6 and 8.

1. Consider $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$, the series behind Model A.

(a) Write down its first term a and its common ratio r.

(b) State which test settles it, and why.

(c) Find the exact sum.

Answer: ____________________

2. Let $a_n = \dfrac{n}{2n+5}$.

(a) Find $\lim_{n\to\infty} a_n$.

(b) State what the nth-term test therefore concludes about $\sum_{n=1}^{\infty} a_n$, and say in one sentence why that conclusion follows.

Answer: ____________________

3. Consider $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3}$.

(a) This is a p-series. Write down p.

(b) Fill in the comparison that decides it: p ______ 1 (write $>$, $<$, or $=$).

(c) State the conclusion and the test that gives it.

Answer: ________________

4. Consider $\displaystyle\sum_{n=1}^{\infty} 6\left(-\tfrac{1}{3}\right)^{n-1}$.

(a) Identify a and r. Note the sign of r — what does a negative ratio do to the terms?

(b) Decide whether the series converges, and find the exact sum if it does.

Answer: ________________

5. Let $a_n = \dfrac{3n^2 + 1}{5n^2 - n}$.

(a) Find $\lim\limits_{n \to \infty} a_n$ exactly.

(b) Apply the nth-term test to $\displaystyle\sum_{n=1}^{\infty} \frac{3n^2 + 1}{5n^2 - n}$ and state the conclusion.

(c) A classmate says the terms of any rational expression shrink to nothing, so the test tells you nothing here. Say in one sentence why that is wrong for this a_n.

Answer: __________________

6. Consider $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$. This one needs both ideas on the sheet.

(a) Write it as $\sum 1/n^p$ and fill in the comparison p ______ 1.

(b) Find $\lim_{n\to\infty} \frac{1}{\sqrt{n}}$.

(c) Your answer to (b) means the nth-term test is *inconclusive* here. Explain what inconclusive means, then give the conclusion the p-series test does reach.

Answer: __________________

7. A geometric model you can see. A ball is dropped from a height of 12 feet. After every bounce it returns to $\frac{3}{4}$ of the height it fell from, and it keeps bouncing forever.

(a) After the first drop, the ball travels up and back down on every bounce, so the bounce distances form the series $\sum_{n=1}^{\infty} 24\left(\frac{3}{4}\right)^n$. Explain in one sentence where the 24 comes from, then find that sum exactly.

(b) Add the original drop to get the total vertical distance the ball travels, in feet.

Answer: ________________ ft

8. Challenge. A classmate writes:

"The terms of $\sum_{n=1}^{\infty} \frac{1}{n}$ shrink to zero. Since the terms go to zero, the series converges."

Model B on the front page is that series. Write a short paragraph that (i) names exactly what the nth-term test can and cannot prove, (ii) gives the correct classification of $\sum 1/n$ with its reason, and (iii) contrasts it with a geometric series whose terms *also* go to zero but which does converge. Name that geometric series.

Answer: ________________